ASP.NET Core Web API Fundamentals

Getting Acquainted with ASP.NET Core



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Architect

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Coming Up



Course structure

Prerequisites, frameworks and tooling

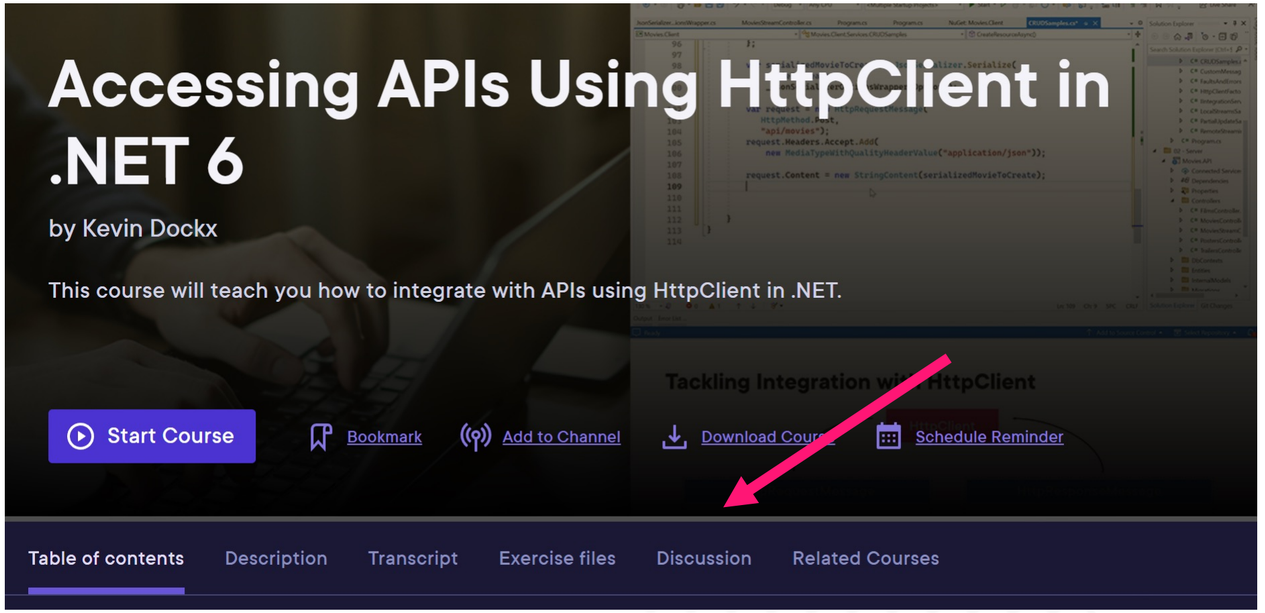
ASP.NET Core: the big picture

Starting an ASP.NET Core project

Middleware and the request pipeline

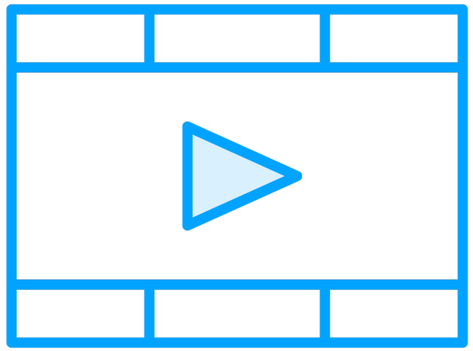
Discussion tab on the course page

BlueSky: @kevindockx.bsky.social
X: @KevinDockx

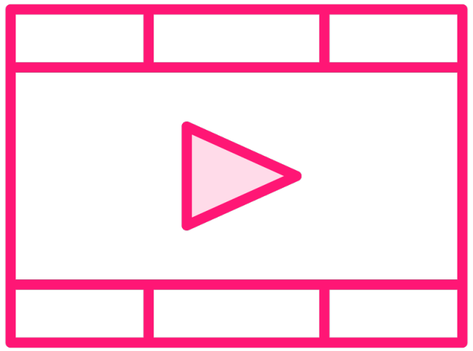


(course shown is one of my other courses, not this one)

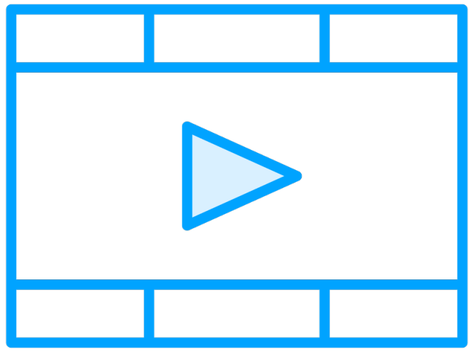
About the Course Structure



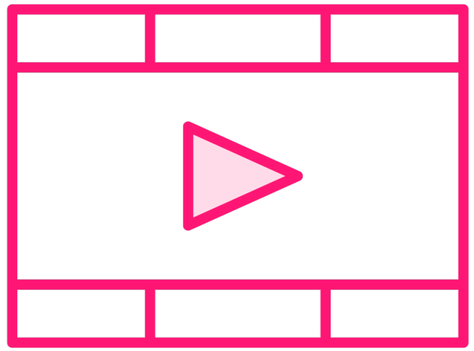
ASP.NET Core Web API Fundamentals



ASP.NET Core Web API Deep Dive



Developing an Asynchronous ASP.NET Core Web API

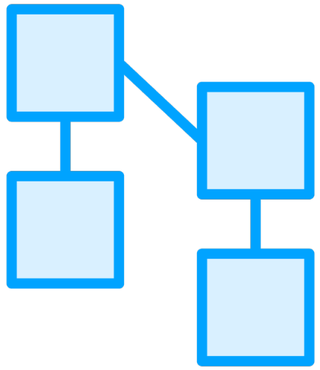


Building ASP.NET Core Minimal APIs

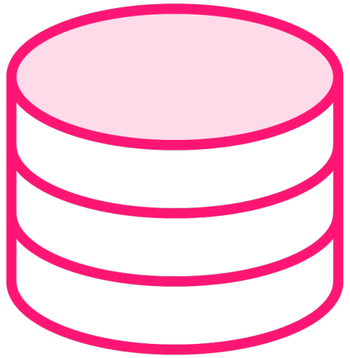
About the Course Structure



From introducing ASP.NET Core...



...over building the API...



...and using Entity Framework Core...



...to common resource concerns, security, versioning, documentation and deployment

Course Prerequisites

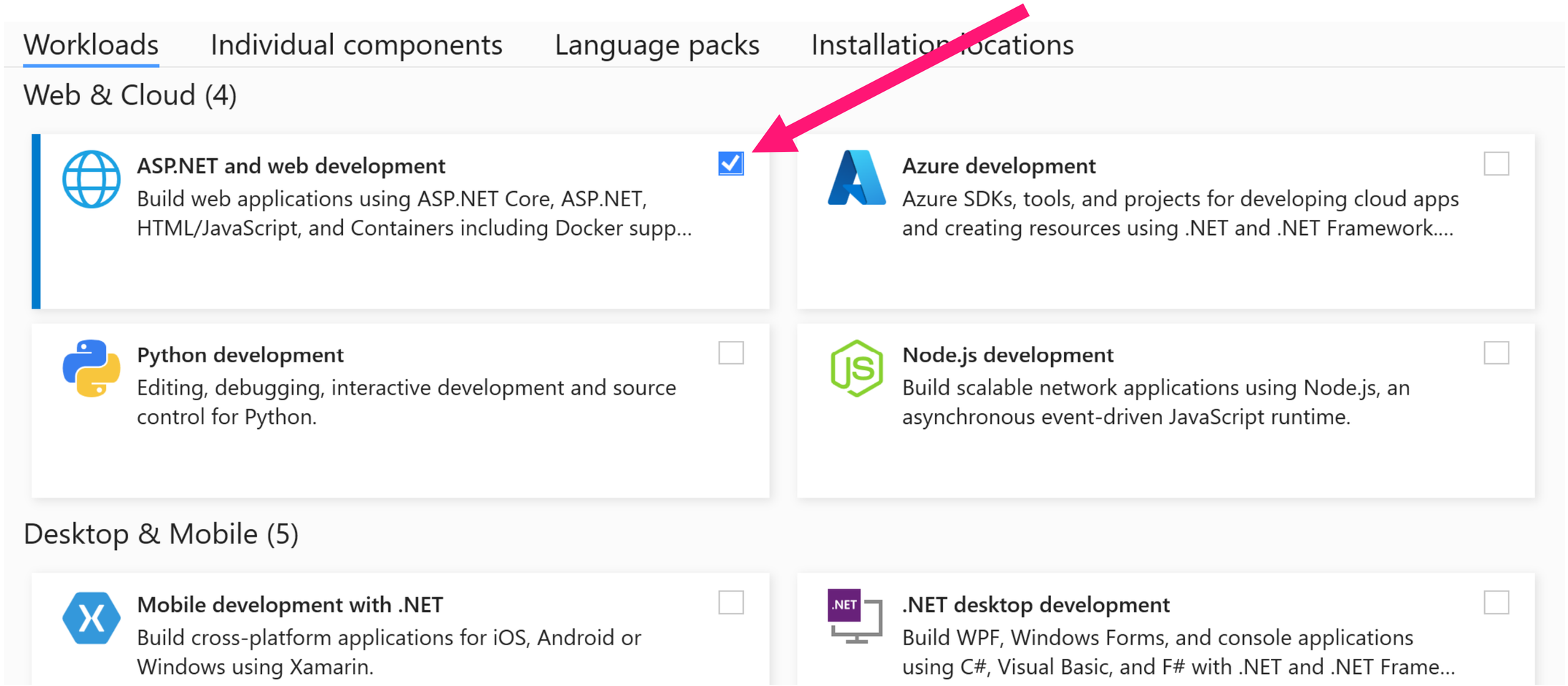
Beginner course, no specific knowledge required regarding

- ASP.NET Core
- Entity Framework Core

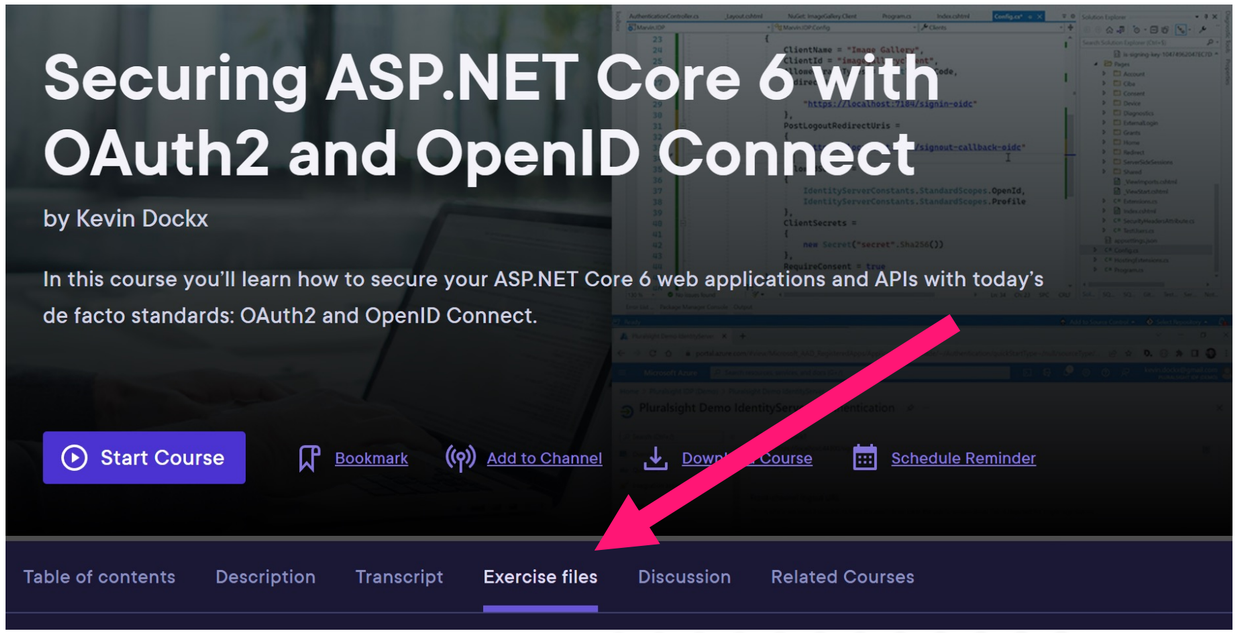
Knowledge of C# is required

- C# 12 is used in this course

Installing Visual Studio



Exercise files tab on the course page



(course shown is one of my other courses, not this one)

ASP.NET Core

A cross-platform, high-performance, open-source framework for building modern, cloud-enabled, internet-connected apps



ASP.NET Core: The Big Picture



Build web apps and services, IoT apps and mobile backends



Develop on Windows, Linux and macOS

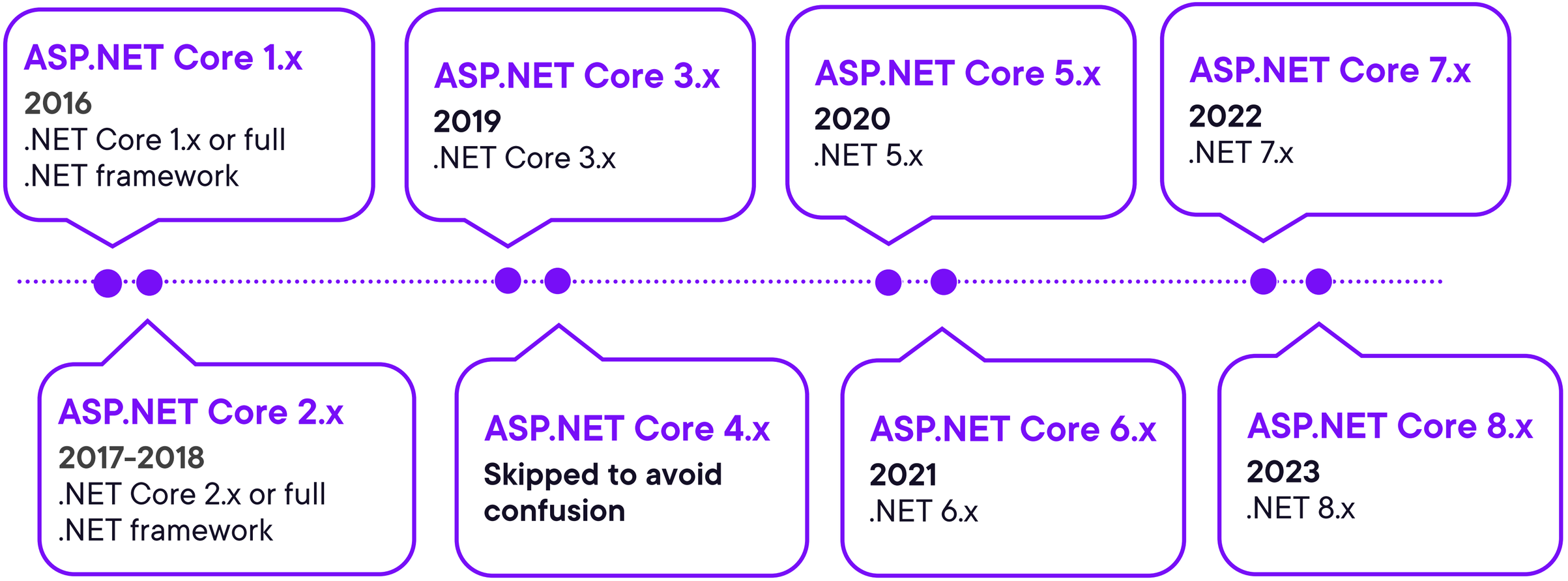


Deploy to the cloud or on-premises

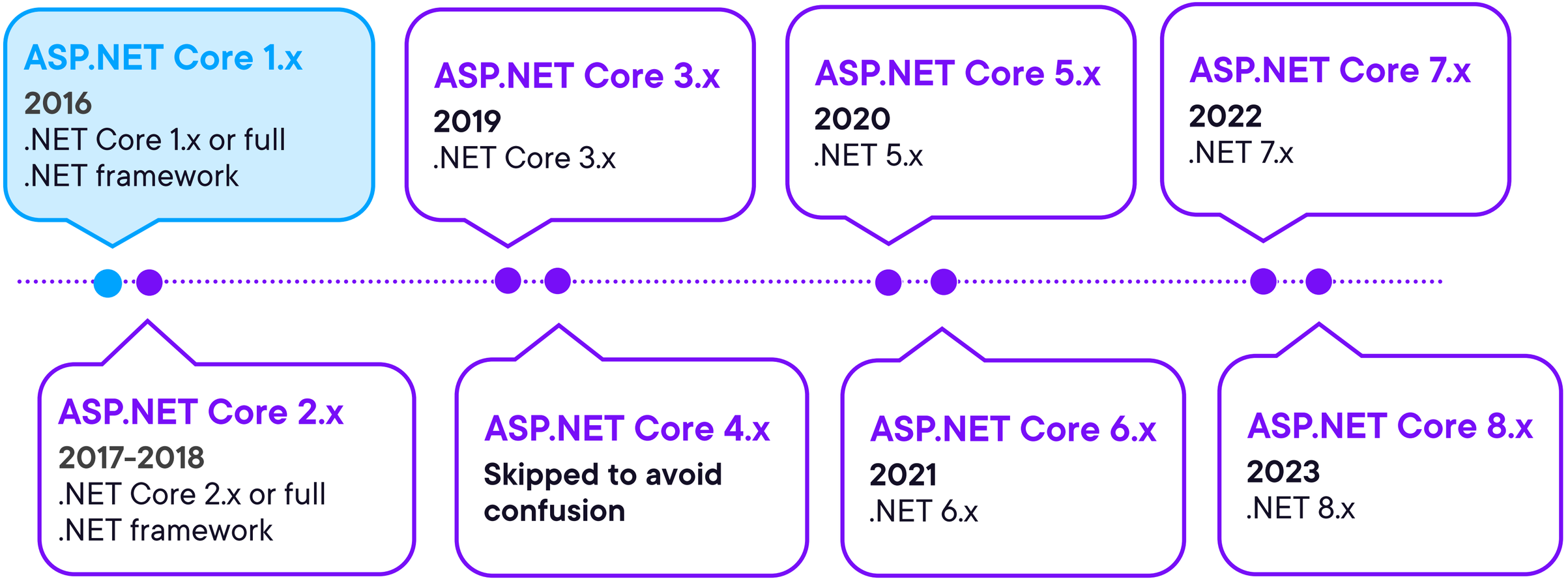


Run on .NET (or .NET Core, depending on the version)

ASP.NET Core History



ASP.NET Core History



ASP.NET Core History

ASP.NET Core 1.x
2016
.NET Core 1.x or full .NET framework

ASP.NET Core 2.x
2017-2018
.NET Core 2.x or full .NET framework

ASP.NET Core 3.x
2019
.NET Core 3.x

ASP.NET Core 4.x
Skipped to avoid confusion

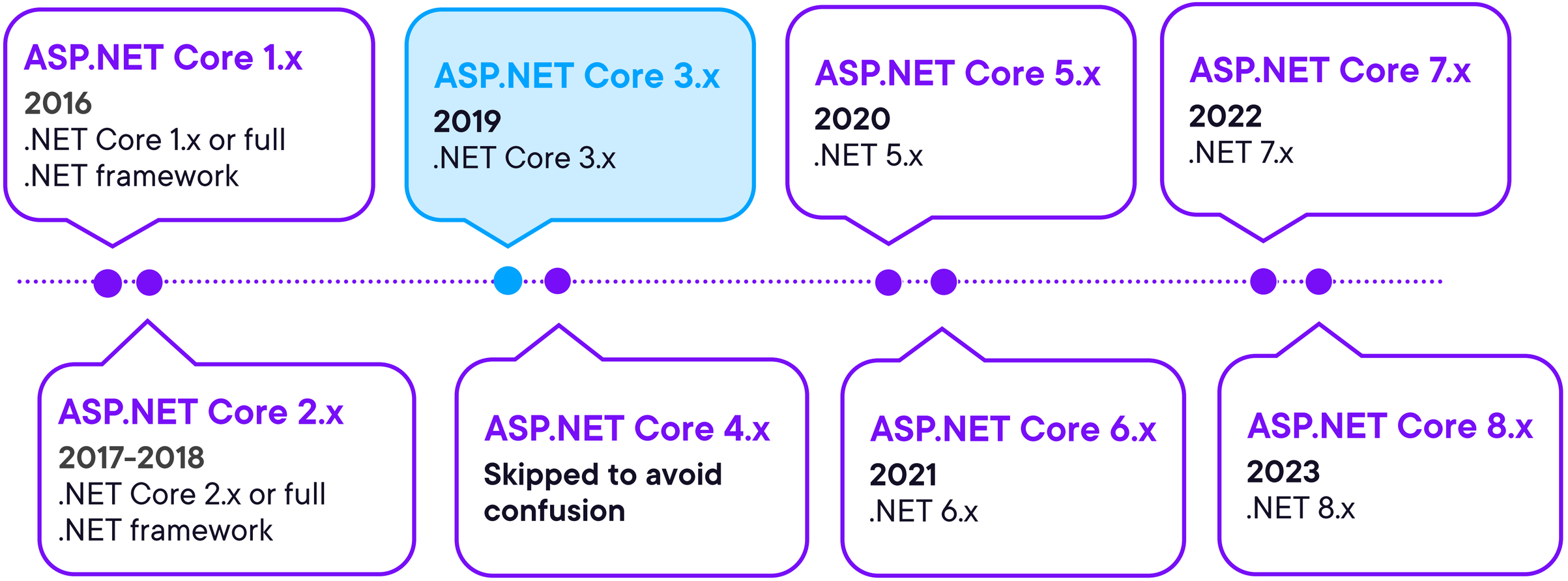
ASP.NET Core 5.x
2020
.NET 5.x

ASP.NET Core 6.x
2021
.NET 6.x

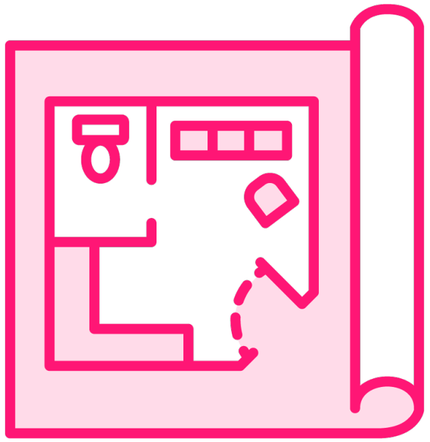
ASP.NET Core 7.x
2022
.NET 7.x

ASP.NET Core 8.x
2023
.NET 8.x

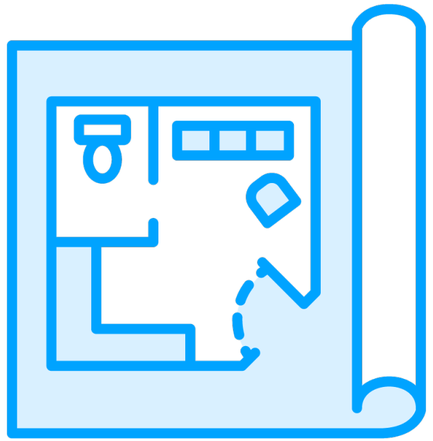
ASP.NET Core History



ASP.NET Core History

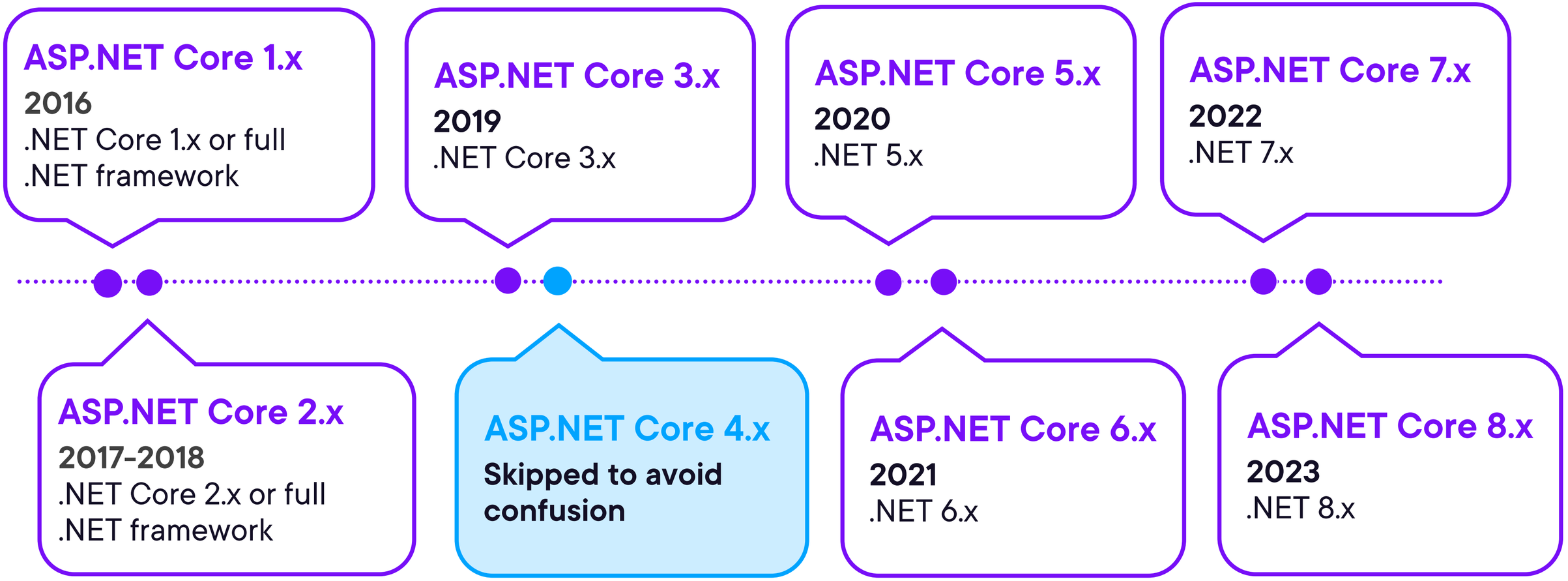


Full .NET framework (4.8)
No new development



.NET Core vNext (“4.x”)
Useful functionality from the full .NET framework ported to increase performance and support cross-platform development

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ASP.NET Core 6.x
2021
.NET 6.x

ASP.NET Core 7.x
2022
.NET 7.x

ASP.NET Core 8.x
2023
.NET 8.x

ASP.NET Core History

ASP.NET Core 9.x

2024

.NET 9.x

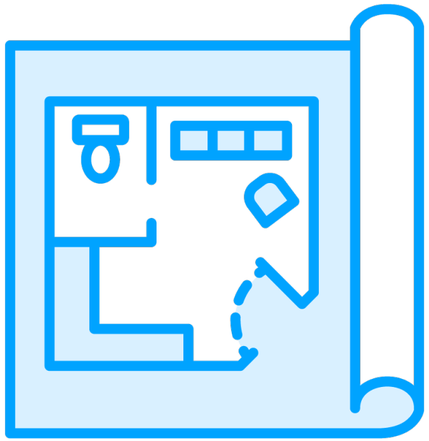
ASP.NET Core History

ASP.NET Core 9.x

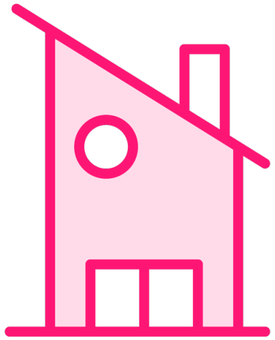
2024

.NET 9.x

Approaches to Building APIs with ASP.NET Core



ASP.NET Core MVC



ASP.NET Core Minimal API

Demo



Creating and running a new ASP.NET Core project

Demo



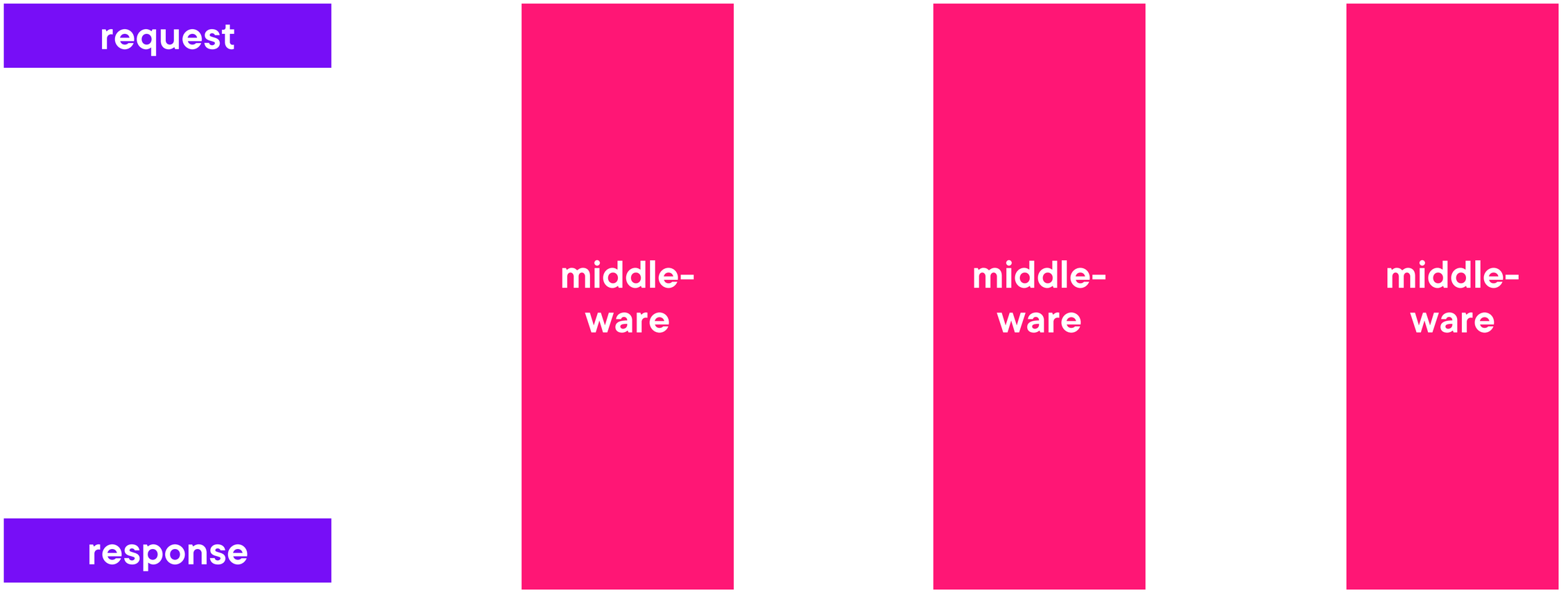
Running an ASP.NET Core project using the CLI

Demo



Inspecting the ASP.NET Core API project code structure

The ASP.NET Core Request Pipeline & Middleware

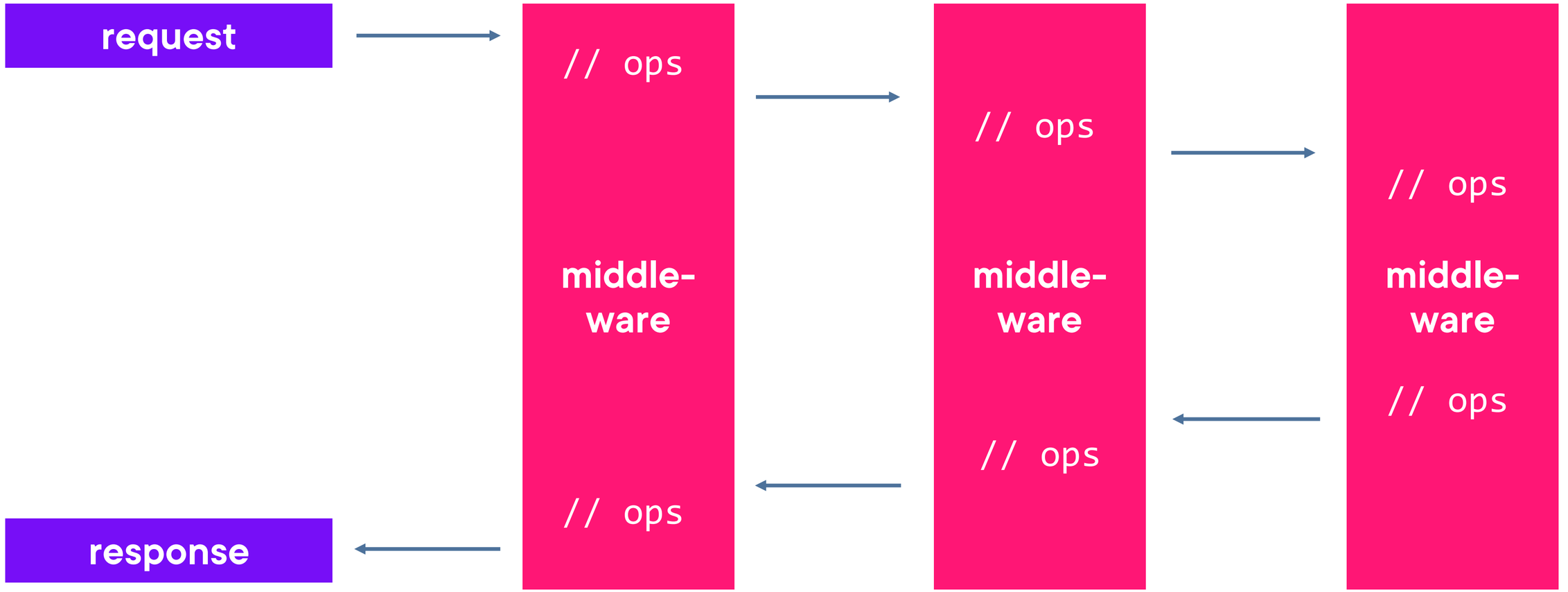


Middleware

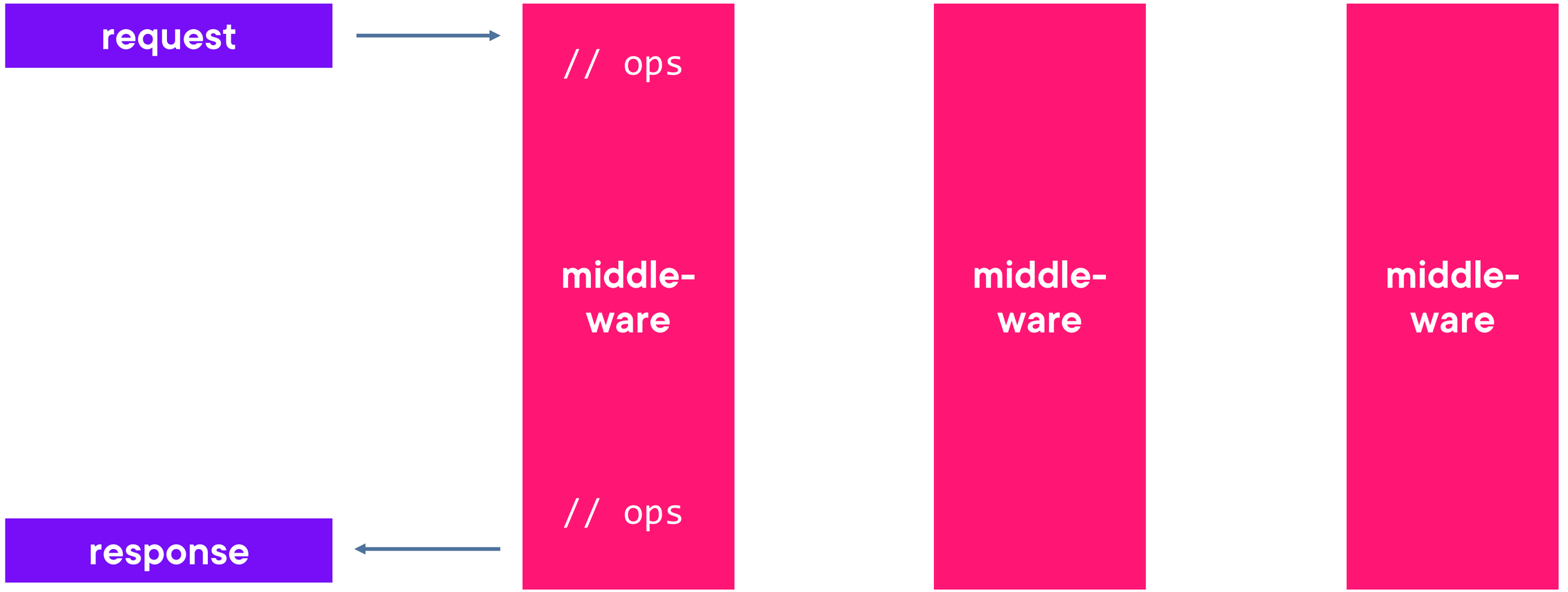
Software components that are assembles into an application pipeline to handle requests and responses



The ASP.NET Core Request Pipeline & Middleware



The ASP.NET Core Request Pipeline & Middleware



Demo



Configuring the ASP.NET Core request pipeline

Demo



Working with different environments

Summary



ASP.NET Core

- Cross-platform
- High-performance
- Open-source
- Modern, cloud-enabled, internet-connected apps

Summary



`Program` class is the starting point of the application

Generated `Main` method is responsible for configuring and running the application

Configure services by adding them to the services collection on the builder (`WebApplicationBuilder`)

Summary



After building the web application via the `WebApplicationBuilder`, configure the request pipeline by adding middleware to it

- Potentially, scope this middleware to different environments

Up Next:

Creating the API and Returning Resources